

Lab 1 warm-up

```
print("Hello World")
```

Hello World

```
# Calculate the volume of a rectangular estuary (length x width x depth)
length = 5000 # metres
width = 800 # metres
depth = 3.5 # metres
volume = length * width * depth
print(f"Estuary volume: {volume} m^3")
```

Estuary volume: 14000000.0 m³

```
# Simulate 10 days of exponential population growth
population = 100
rate = 0.05
history = [population]

for day in range(10):
    population = population + rate * population
    history.append(round(population, 1))

print(history)
```

[100, 105.0, 110.2, 115.8, 121.6, 127.6, 134.0, 140.7, 147.7, 155.1, 162.9]

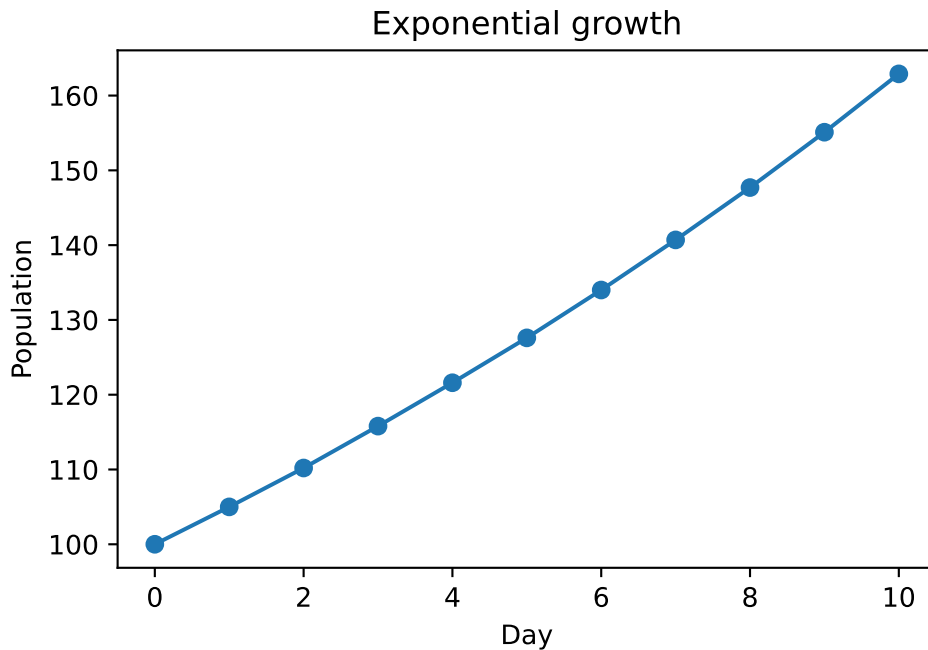
```
# Write a function that converts temperature from Celsius to Fahrenheit
def celsius_to_fahrenheit(c):
    return c * 9/5 + 32

# Test it
for temp in [0, 15, 25, 37]:
    print(f"{temp}°C = {celsius_to_fahrenheit(temp)}°F")
```

0°C = 32.0°F
 15°C = 59.0°F
 25°C = 77.0°F
 37°C = 98.6°F

```
import matplotlib.pyplot as plt

days = list(range(len(history)))
plt.plot(days, history, "o-")
plt.xlabel("Day")
plt.ylabel("Population")
plt.title("Exponential growth")
plt.show()
```



```

import matplotlib.pyplot as plt

concentration = 100.0
decay_rate = 0.01
dt = 1.0
history = [concentration]

for day in range(60):
    concentration = concentration - decay_rate * concentration * dt
    history.append(concentration)

plt.plot(history)
plt.xlabel("Day")
plt.ylabel("Concentration (mg/L)")
plt.title("Tracer decay in a lake")
plt.show()

# Change decay_rate to 0.1. Does the tracer disappear faster or slower? Ans: The tracer disappears faster.
# Change decay_rate to 0.01. What happens? Ans: The decay curve becomes flatter, meaning the tracer stays longer.
# In one sentence: what does the decay rate control? Ans: The decay rate controls how quickly the tracer disappears.

```

